

API Readiness & Load-Test Report

GravelKing Pro / Morris Law Kernel v3.5 • Prepared August 4, 2026 for the Joel integration meeting

1,000/1,000
0

health checks returned
HTTP 200

500/500

MLK v3 requests returned
HTTP 200

74/74

automated API checks
passed

3 lanes

current mastering
concurrency cap

Executive readout

The core is healthy and measured.

The public API is live. Controlled tests passed across health, MLK processing, malformed input, and missing routes.

GO

Core health and load evidence

HOLD

Partner credential handoff

The partner handoff gates are simple: publish **/api/v1/ingest**, hard-gate **x-api-key**, and run one real production smoke test before issuing a credential.

What we tested

Controlled traffic results • clear handling of valid, invalid, and unknown requests

Test	Observed result	Meaning
Health endpoint	1,000 / 1,000 HTTP 200	No failed health requests in the controlled run.
MLK v3 processing	500 / 500 HTTP 200	Kernel processing route stayed available under the test load.
Raw processing baseline	1,000 / 1,000 HTTP 200	Baseline API request handling remained stable.
Invalid payloads	200 / 200 HTTP 400	Bad requests were rejected cleanly, not as server crashes.
Missing routes	200 / 200 HTTP 404	Unknown paths failed predictably.

Test	Observed result	Meaning
After-load health	HTTP 200	The service remained healthy after traffic stopped.

Automated API verification

74 passed checks. 0 failures. 17 technical-brief checks • 16 studio/MLK checks • 20 pricing checks • 21 download-security checks.

Partner stress result

The system applies deliberate back-pressure instead of crashing under excess concurrent mastering work

3
Requests
processed

27
Clean HTTP
503
responses

3
Active
mastering
lanes

Interpretation: the service refused excess simultaneous work cleanly. This protects CPU and memory. It is not a crash.

Honest boundary: a three-lane synchronous service is not yet a durable 500-track batch queue. It needs job IDs, retries, idempotency, status, and callbacks before a production 500-track promise.

How the IP diff / verification works

The system does not use Diffie–Hellman. It uses a split-proof record tied to the audio bytes.

How we know what changed

A certified track receives a **SHA-256 content fingerprint** and two linked anchors. **Anchor A** is embedded in lossless WAV/PCM audio. **Anchor B** stays on the server. An **HMAC-SHA256** binding proves the two anchors belong to the same certificate.

Step	What happens	Result
1. Read	Extract the embedded Anchor A and certificate ID from the submitted signal.	If absent: NO_WATERMARK.
2. Hash	Hash the received bytes and compare them with the stored certificate record.	A changed file produces a different fingerprint.

3. Bind	Recompute the server-side HMAC using the hidden Anchor B and extracted Anchor A.	Only the server can produce the valid binding.
4. Verdict	Return a structured result for the partner or investigator.	INTACT, TAMPERED, or NO_WATERMARK.

A certificate validates the issued signal and stored provenance chain. It is not, by itself, a court ruling or a guarantee of legal ownership.

Measured speed vs. capacity plan

Keep the observed evidence separate from the projected partner pilot

Measured now

109× realtime was measured as a separate DSP benchmark on commodity CPU hardware. That is not an end-to-end API service-level agreement. The current API has **3 active mastering lanes**.

Projected pilot

Planning case	Illustrative estimate	Status
500 tracks at 3 concurrent lanes	At an 8–45 second mastering range: roughly 22–125 minutes, before retries, upload time, and queue overhead.	Projection only — validate with Joel's real track mix.

Controlled partner pilot	Start with 10 tracks, then 50, then 100; observe p50/p95 duration, errors, CPU, memory, and queue depth.	Recommended next step.
500-track production batch	Use a queue, durable job IDs, retries, idempotency, status polling, and optional callbacks before promising this.	Not ready today.

Joel handoff checklist

Use this page as the meeting close: agree on the pilot, then finish the production handoff gates.

State	Meeting item	What to say / confirm
✓	Evidence	Show the 1,000/1,000 health result, 500/500 MLK result, and 74/74 API checks.
✓	Explain the diff	Walk through Anchor A in the audio, Anchor B on the server, SHA-256, HMAC, and the three verdicts.
■	Publish	Publish the source containing POST /api/v1/ingest and POST /api/v1/verify.
■	Secure	Make x-api-key a hard gate and verify no-key / wrong-key requests fail before audio processing.

State	Meeting item	What to say / confirm
■	Smoke test	Run one real production audio file with a valid partner key; confirm output headers and certificate behavior.
■	Pilot limits	Agree on 10 → 50 → 100 tracks first; measure p50/p95 latency, errors, CPU, memory, and queue depth.
■	Scale plan	Add queueing, retries, idempotency, job status, and callbacks before a 500-track production promise.

Suggested opening: “The core is already tested: 1,000 of 1,000 health requests, 500 of 500 MLK processing requests, and 74 of 74 automated checks passed. What I want to agree with you today is the controlled partner pilot and the final production handoff gates—not pretend that a three-lane synchronous service is already a 500-track queue.”

Source note: results are from controlled workspace/API tests and production endpoint probes available on August 4, 2026. Projection figures are explicitly labeled estimates and must be re-measured with the partner's real workload.